TOYSOUND-01

Low-Power Digital Sound Controller ASIC
COB ASIC Laboratory — Project 01

COB ASIC Lab

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1 Project Status

Phase	Status
SPECIFICATION	[COMPLETE]
ARCHITECTURE	[COMPLETE]
RTL	[COMPLETE]
SIMULATION	[COMPLETE]
FPGA	[PENDING]
ASIC SYNTHESIS	[PENDING]
PHYSICAL DESIGN	[PENDING]
DRC / LVS	[PENDING]
GDSII	[PENDING]
TAPEOUT	[PENDING]
COB / PCB	[PENDING]

2 Executive Summary

TOYSOUND-01 is a minimal digital ASIC designed for a toy sound module. A single button cycles through four square-wave melodies. Target process: **SCL 1.2 μm CMOS (C1D)**. The chip is intended for Chip-On-Board (COB) packaging on a minimal PCB with a battery, a single button, and a piezo sounder.

```
[BUTTON] --> [DEBOUNCE] --> [FSM / CONTROLLER]
                                    |
                            [SOUND SEQUENCER]
                                    |
                          [FREQUENCY DIVIDER / TONE GEN]
                                    |
                              [SOUND_OUT pin]
                                    |
                           external driver / sounder
```

3 Digital Architecture

3.1 Modules

Module	Function
`toysound_pkg.sv`	Global constants: note frequencies, durations
`toysound_top.sv`	Top wrapper; exposes 6 physical pads
`button_debounce.sv`	Filters mechanical bounce with a stable counter
`sound_controller.sv`	4-state FSM; cycles song index on each press
`sound_sequencer.sv`	Case-statement ROM; steps through note table
`timer.sv`	Countdown register; fires done pulse on zero
`tone_generator.sv`	Toggle pin at note frequency (PWM square wave)

3.2 FSM States

```
IDLE --> (button press) --> LOAD_SOUND --> PLAYING --> DONE --> IDLE
```

3.3 Pad / Pin Specification

Pin	Name	Dir	Function
1	VDD	Power	Supply voltage
2	GND	Power	Ground
3	CLK	Input	System clock (1 MHz target)
4	RESET_N	Input	Active-low async reset
5	BTN	Input	Next-sound button
6	SOUND_OUT	Output	Square-wave to driver

Note: Direct speaker drive capability is PDK-dependent and TBD until verified against SCL 1.2 μm I/O specifications.

4 Simulation

4.1 Tool

Icarus Verilog 12 (`iverilog` + `vvp`). Waveforms viewed in GTKWave.

4.2 Simulation Flow

```
RTL (.sv)  -->  iverilog (compile)  -->  .vvp  -->  vvp (run)  -->  .vcd
                                                                      |
                                                                  GTKWave
```

4.3 Test Results

Test	Description	Result
T01	Reset clears all outputs	PASS
T02	Button press 1 plays sound 1	PASS
T03	Button press 2 plays sound 2	PASS
T04	Button press 3 plays sound 3	PASS
T05	Button press 4 plays sound 4	PASS
T06	Button press 5 wraps to sound 1	PASS
T07	Bounce suppressed (no false event)	PASS
T08	Button ignored during playback	PASS
T09	Reset during playback recovers safely	PASS

5 ASIC Synthesis

Tool: Yosys **PDK:** SCL 1.2 μm C1D Standard Cell Library

```
RTL (.sv)  -->  yosys -c synth.tcl  -->  syn/toysound_netlist.v
                                         syn/toysound_stats.txt
```

STATUS: NOT YET RUN

6 Physical Design

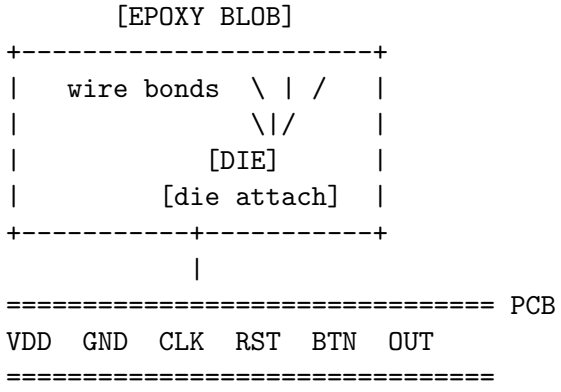
Tool: OpenROAD (via WSL2/Docker) **STATUS: PENDING**

```
netlist  -->  Floorplan  -->  Placement  -->  CTS  -->  Routing  -->  impl/toysound.gds
```

7 SCL 1.2 μm PDK

Component	Used in
Liberty (.lib)	Synthesis (Yosys), STA
LEF	Physical design (OpenROAD)
GDS	Final GDSII merge
DRC rule deck	Design rule check (signoff)
LVS netlist	Layout vs schematic (signoff)

8 COB Packaging Concept



STATUS: AWAITING FABRICATION

9 File Index

File	Type	Purpose
01_rtl/core/sound_controller.sv	RTL	SystemVerilog source
01_rtl/core/sound_sequencer.sv	RTL	SystemVerilog source
01_rtl/core/timer.sv	RTL	SystemVerilog source
01_rtl/core/tone_generator.sv	RTL	SystemVerilog source
01_rtl/peripherals/button_debouncer.sv	RTL	SystemVerilog source
01_rtl/pkg/toysound_pkg.sv	RTL	SystemVerilog source
01_rtl/top/toysound_top.sv	RTL	SystemVerilog source
02_tb/toysound_top_tb.sv	Testbench	Verification
03_sim/waveforms/toysound_top.vcd	Waveform	Simulation output
03_sim/toysound.vvp	Compiled	Icarus compiled sim